

Answer key - Midterm - Digital Electronics - 2025 (Sep)

$$1(a). F(A,B,C) = (AB)'(AC)' + \underbrace{A'B'C}_{X+0=X}$$

$$= \underbrace{(AB)'(AC)'}_{\text{Demorgan's law}} + A'B'C$$

$$= (AB+AC)' + A'B'C$$

$$= \underbrace{(A(B+C))'}_{\text{Demorgan's}} + A'B'C$$

$$= \underbrace{A' + (B+C)'}_{\text{Absorption law}} + \underbrace{A'B'C}_{X+XY=X}$$

$$= A' + (B+C)'$$

$$= \underline{A' + B'C'}$$

$$1(b) F(x,y,z) = xy' + y'z' + x'z$$

$$| \text{ Dual} \rightarrow (x+y')(y'+z')(x'+z)$$

$$| \text{ Complement} \rightarrow (xy' + y'z' + x'z)'$$

$$= (xy')' \cdot (y'z')' \cdot (x'z)'$$

$$= (x'+y) \cdot (y+z) \cdot (x+z')$$

Dual & Complement are different operations

$$\text{In terms } \text{dual}(F') = (\text{dual}(F))'$$

→ Complement uses De-morgan's laws, dual is swapping of operands.

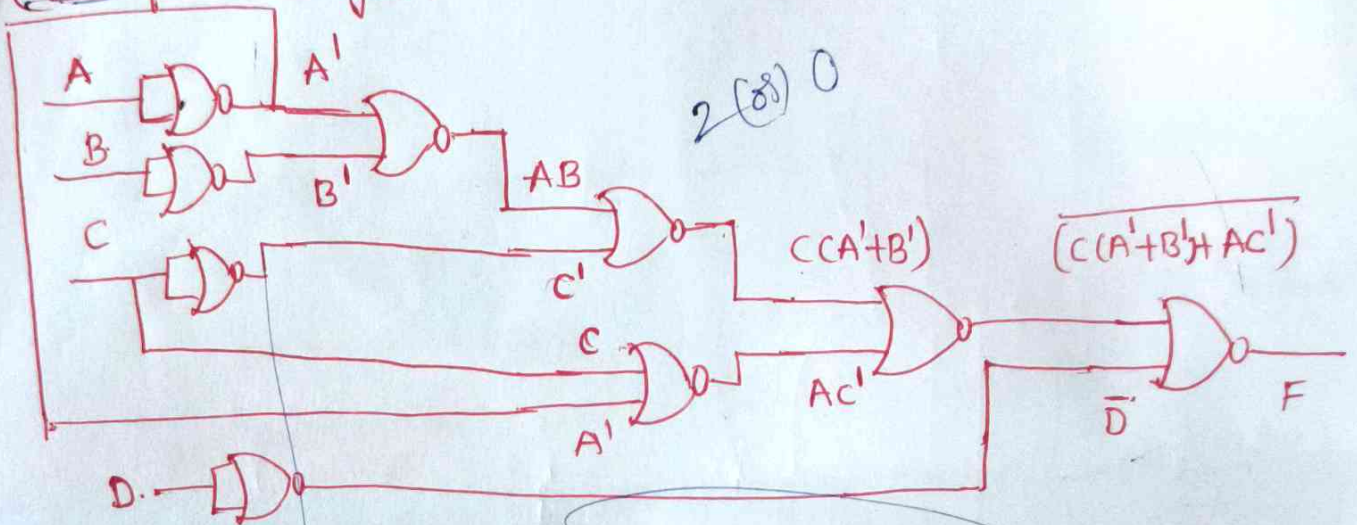
Top Branch $A'B' \rightarrow \text{parallel}$
 $C \rightarrow \text{series} \} \rightarrow (A'+B')C$

Bottom branch $A, c' \rightarrow \text{series} \rightarrow AC'$

Both (Top || Bottom) in series with D $\rightarrow D((A'+B')C + AC')$

$$F = D(C(A'+B') + AC')$$

(2-input - NOR gate)

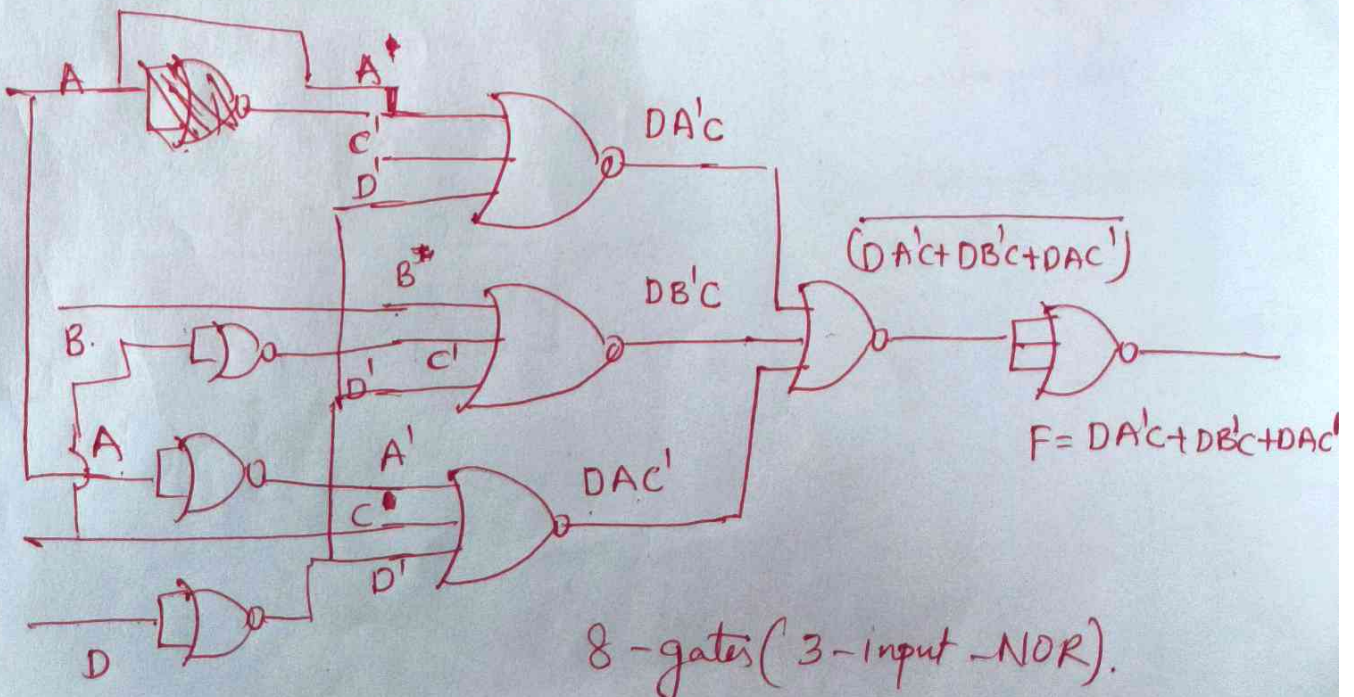


2 (8) 0

3-input - NOR gates

9 - NOR gates (2-input)

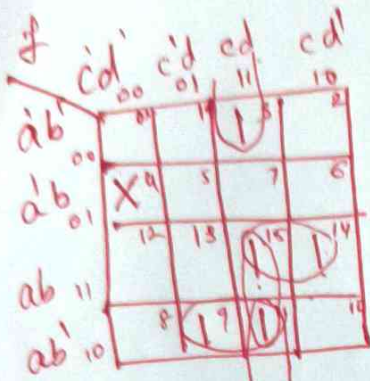
$$F = DA'C + DB'C + DAC'$$



8 - gates (3-input - NOR).

$$f(a,b,c,d) = \sum m(3,9,11,14,15) + d(4)$$

$$g(a,b,c,d) = \sum m(0,2,3,9,11,12,13) + d(14)$$



$$g_1 = ab'cd + ab'cd = ab'd$$

$$g_2 = abcd + ab'cd = acd$$

$$g_3 = abcd + abcd' = abc$$

$$g_4 = ab'cd + a'b'cd = b'cd$$

$$f = g_1 + g_2 + g_3 + g_4 \quad \text{Redundant term}$$

$$= ab'd + acd + abc + b'cd$$

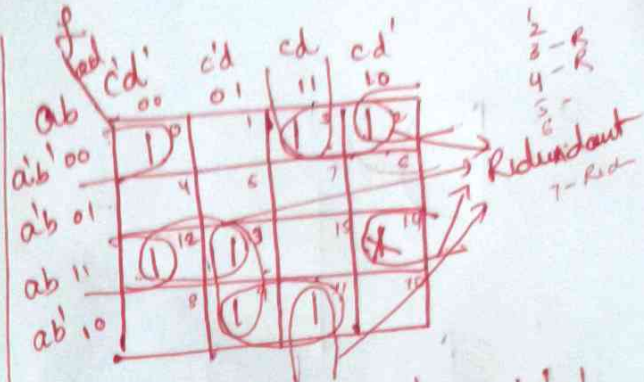
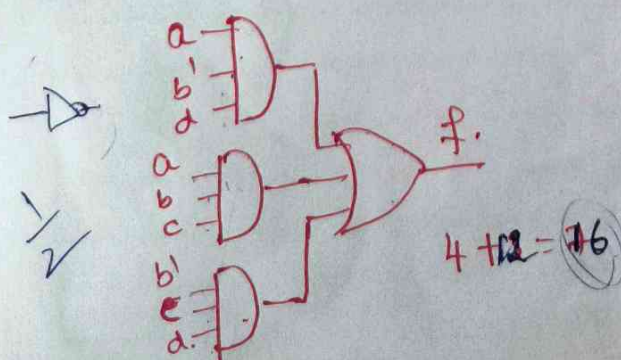
$$= (a+c)b'd + ac(b+d)$$

$$= ab'd + abcd + ab'cd + abc + b'cd$$

Absorption law $\leftarrow (a+ac)b'd + abc(d+1) + b'cd$

$$= ab'd + abc + b'cd$$

$$f = ab'd + abc + b'cd$$



$$g_1 = ab'cd + ab'cd = ab'd$$

$$g_2 = ab'cd + ab'cd = b'cd$$

$$g_3 = ab'cd + ab'cd = a'b'c \text{ (Redundant)}$$

$$g_4 = abcd + abcd' = abd$$

$$g_5 = abcd + abcd = abc \text{ (Redundant)}$$

$$g_6 = abcd + abcd = acd \text{ (Redundant)}$$

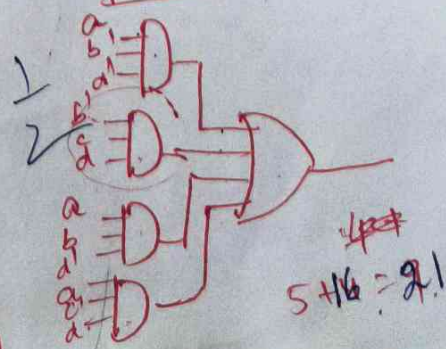
$$g_7 = ab'cd + ab'cd = ab'd$$

$$g = g_1 + g_2 + g_3 + g_4 + g_5 + g_6 + g_7$$

(minimized O/P)

$$g = abc + abd + acd + b'cd$$

$$g = ab'd + b'cd + abd + ab'd$$

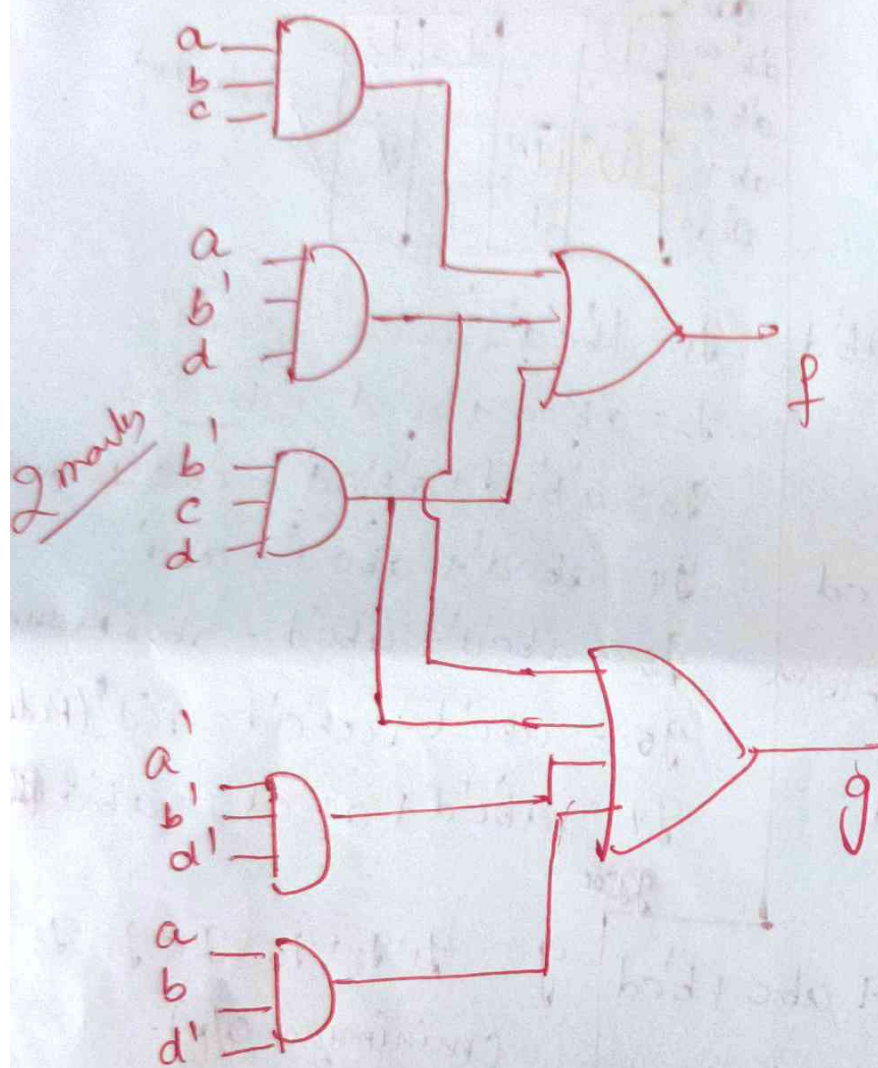


Total Cost =

$$7 + 16 = 23$$

(37)

(ii)



$$22 \text{ (wires)} + 7 \text{ gates} = 29 \text{ gates}$$

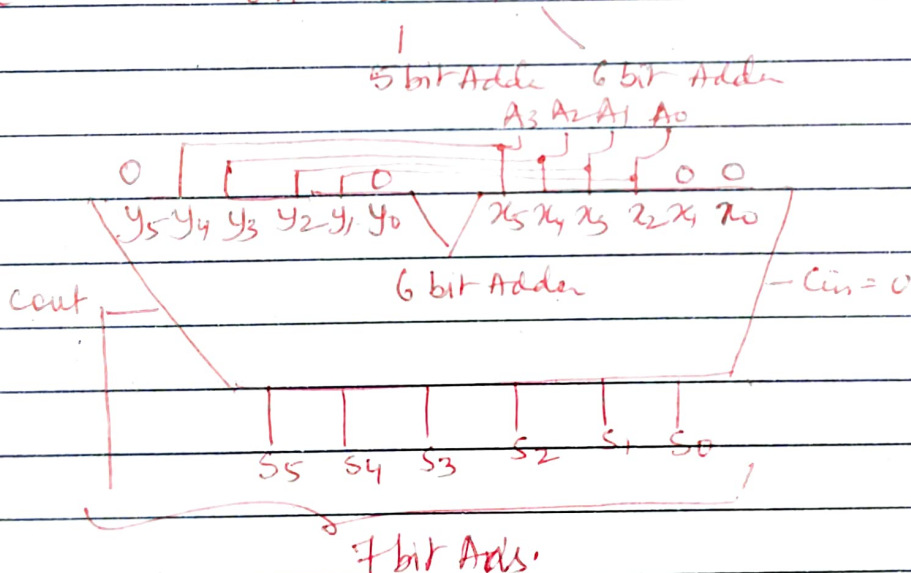


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Q2b → 3 bit Binary to Gray code converter

B_2	B_1	B_0	G_2	G_1	G_0	$G_2 = B_2$
0	0	0	0	0	0	$G_1 = B_2 \oplus B_1$
0	0	1	0	0	1	$G_0 = B_1 \oplus B_0$
0	1	0	0	1	1	
0	1	1	0	1	0	
1	0	0	1	1	0	
1	0	1	1	1	1	
1	1	0	1	0	1	
1	1	1	1	0	0	

Q3a. $P = 6A = 2A + 4A$



(5M)

$A+A$
 $A+A$
 $P+A$
3M

for SOP-2M

TIT-2M-

S-1 → 1 (reduced for)

C-1 → 1



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15M

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Q.3b

A B Cin

Sum Cout

Sum

AB

A+B

A+B

A+B

A+B

A+B

0 0 0

0 0

Cin

00

01

11

10

0 0 1

1 0

C

00

12

03

14

0 1 0

1 0

C

11

03

17

05

0 1 1

0 1

AB

Cout

00

01

11

10

1 0 0

1 0

Cin

00

01

11

10

1 0 1

0 1

0

00

02

16

04

1 1 0

0 1

1

01

15

17

15

1 1 1

1 1

13

$$\text{Sum} = (A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+B+\bar{C})$$

$$\text{Cout} = (A+B)(A+C)(A+C) - (1M)$$

For SOP-3M

A=0 (A)

A=1 (A)

DE	00	01	11	10
00	0 ⁰	0 ⁴	0 ¹²	0 ⁸
01	1 ¹	1 ⁵	1 ¹³	1 ⁹
11	1 ³	0 ⁷	1 ¹⁵	1 ¹¹
10	0 ²	1 ⁶	1 ¹⁴	0 ¹⁰

DE	00	01	11	10
00	0 ¹⁶	0 ²⁰	0 ²⁸	0 ²⁴
01	1 ¹⁷	1 ²¹	1 ²⁹	0 ²⁵
11	1 ¹⁹	0 ²³	1 ³¹	0 ²⁷
10	0 ¹⁸	1 ²²	1 ³⁰	0 ²⁶

$$(D+E)$$

$$(C+E)$$

$$(B+\bar{C}+\bar{D}+\bar{E})$$

$$(\bar{A}+\bar{B}+C)$$

Kmap-(2M)

$$F = (D+E)(C+E)(\bar{A}+\bar{B}+C)(B+\bar{C}+\bar{D}+\bar{E}) - (4M)$$

All are EPI's. — (1M) only if all are correct

Initials of Staff-in-charge & Date :

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$$F(A, B, C, D) = \bar{B}D + \bar{A}D + BD = D$$
$$= 1$$

$$= \Pi_m(0, 2, 4, 6, 8, 10, 12, 14)$$

CD \ AB	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

(2M)

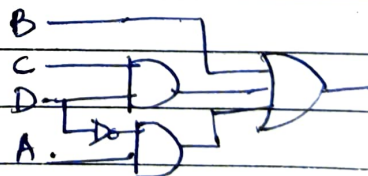
(1M)

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

If don't care not used - 6 marks
NO kmap - 4 marks

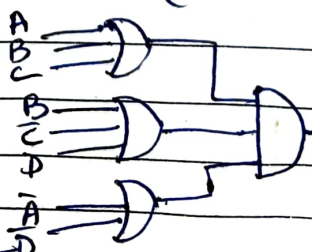
CD \ AB	00	01	11	10
00	0	1	X	1
01	0	1	X	0
11	1	1	X	X
10	0	1	X	X

$$F = B + CD + A\bar{D}$$

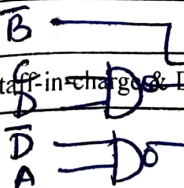


10 cost
✓

$$F = (A + B + C)(B + \bar{C} + D)(A + \bar{D})$$



4 + 11 = 15

Initials of Staff-in-charge & Date: 

10 cost

(2M)